Expressions and operator order

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Teach precedence through explicit groupings and concrete assumptions.

Supply complete scripts, isolate zero-divisor errors and assess an explained fresh expression after exposure checks.

Keep one habit: Ask which operation groups first, what quantity it represents and whether the inputs satisfy its interpretation.

MAIN LESSON · SECTION 1

Prepare the connected bridge lesson

> Check prior exposure and keep execution practice distinct from paper reasoning.

Goal and main plan

Primary target: Evaluate numeric **expression**s using Python’s grouping rules

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Retrieval and prerequisite guidance

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish integer, floating-point, text and Boolean values

Reference example, not the only acceptable wording: 12 is int; 1.5 is float; "12" is str; False is bool; "False" is str. Quotes mark text even when the characters spell a Boolean name.

Teach grouping through meaning before mnemonic rules. Multiplication and division share **precedence**; addition and subtraction share **precedence**.

This lesson deliberately limits whole-group examples to nonnegative integers and positive divisors.

Reference: Python tutorial arithmetic, https://docs.python.org/3/tutorial/introduction.html, and **expression precedence**, https://docs.python.org/3/reference/**expression**s.html#**operator**-**precedence**.

Materials and readiness

Python 3 editor with **expression**s.py and separate source/**output** areas

Counters or sketches of packs with separate loose items

Blank **expression**, grouping, expected and observed columns

Numeric addition gives 10; text concatenation gives the string 82. This week uses numeric values.

End goal

Trace **expression**s and **operator** order and explain the changed program before running it.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Values variables and types

Outcomes

Evaluate numeric **expression**s using Python's grouping rules

Use **parentheses** to express the intended calculation

Distinguish ordinary division, whole groups and a **remainder**
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the written grouping and the arithmetic separately

> Label the packing, dimensionless picture and same-precedence records.

State the counting model and Python workflow

Three packs each contain eight counters, with two more loose counters. First count the packed counters, then add the loose ones. A program needs that structure, not just the right numbers.

This week you will predict **expression**s, make grouping visible and check division against a physical interpretation.

Use numeric values in a fresh Python 3 script called **expression**s.py. Keep expected and observed results separate as before. Multiplication and division group more tightly than addition and subtraction.

Parentheses let you state a different grouping. A correct calculation still depends on correct quantities and units.

The script displays 26. Replacing the total **expression** with packs * (per_pack + loose) displays 30, because it adds two to every pack. **Parentheses** are not decoration: they can change the represented situation.

Write a brief phrase for each group before running.

Trace the same-precedence reference

```
print(20 - 6 - 2)
print(20 - (6 - 2))
print(12 / 3 * 2)
```

The first line groups subtraction from left to right: (20 - 6) - 2 gives 12. The second gives 16. Multiplication and division share a **precedence** level, so 12 / 3 * 2 groups as (12 / 3) * 2 and displays 8.0, not 2.0.

Ordinary / division produces a float, even when the result is mathematically whole.

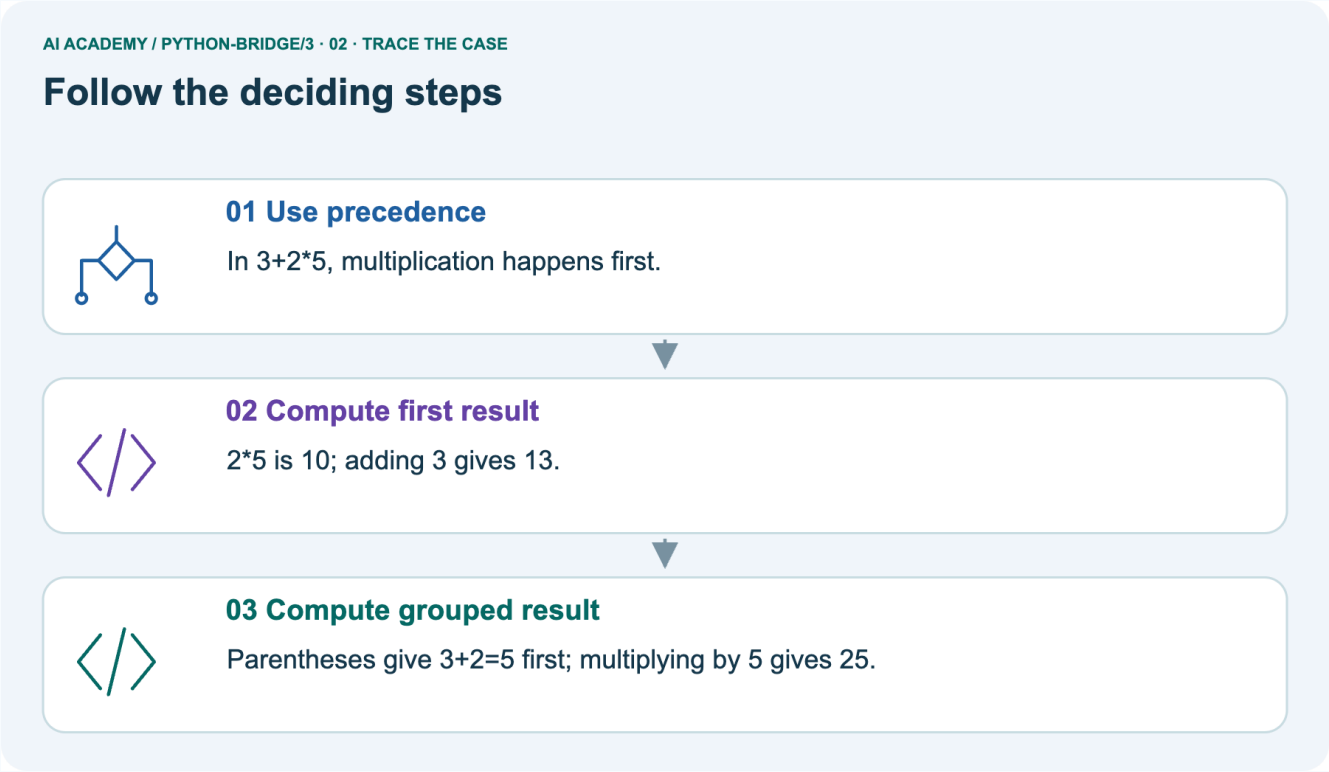
```
print(16 - 5 - 3)
print(16 - (5 - 3))
print(18 / 3 * 2)
```

Read the dimensionless pair

Compare two dimensionless arithmetic **expression**s: adding 3 after 2*5 versus multiplying the whole 3+2 group by 5. Match each group to a written model before using it for a real counting task.

Python 3: print(3+2*5); print((3+2)*5). **Input**s are dimensionless for this arithmetic exercise.

Trace precedence and parentheses



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Use precedence	In 3+2*5, multiplication happens first.
2. Compute first result	2*5 is 10; adding 3 gives 13.
3. Compute grouped result	Parentheses give 3+2=5 first; multiplying by 5 gives 25.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

```
print(3 + 2 * 5)
print((3 + 2) * 5)
```

Predict the multiplier change

Change: Replace the final multiplier 5 with 4 in both **expression**s.

Prediction: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

State model/domain limits

Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional guided key

In 18 - 4 * 3 multiplication groups first, giving 18 - 12 = 6. In (18 - 4) * 3 the **parentheses** give 14 * 3 = 42.

Exit: 23 // 4 gives 5 trays and 23 % 4 gives 3 tiles; 5 * 4 + 3 = 23. / gives 5.75, a ratio rather than both requested whole-number counts.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Expression	Code evaluated to produce a value
Operator	A symbol such as + or * that specifies an operation
Precedence	Rules that determine how an expression is grouped
Parentheses	Symbols that make a grouping explicit
Remainder	The amount left after forming whole groups

Two expressions with different meanings

EXPRESSION	FIRST CALCULATION	FINAL VALUE
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Explain why 18 - 4 * 3 gives 6 while (18 - 4) * 3 gives 42. Which operation is grouped first in each expression?**

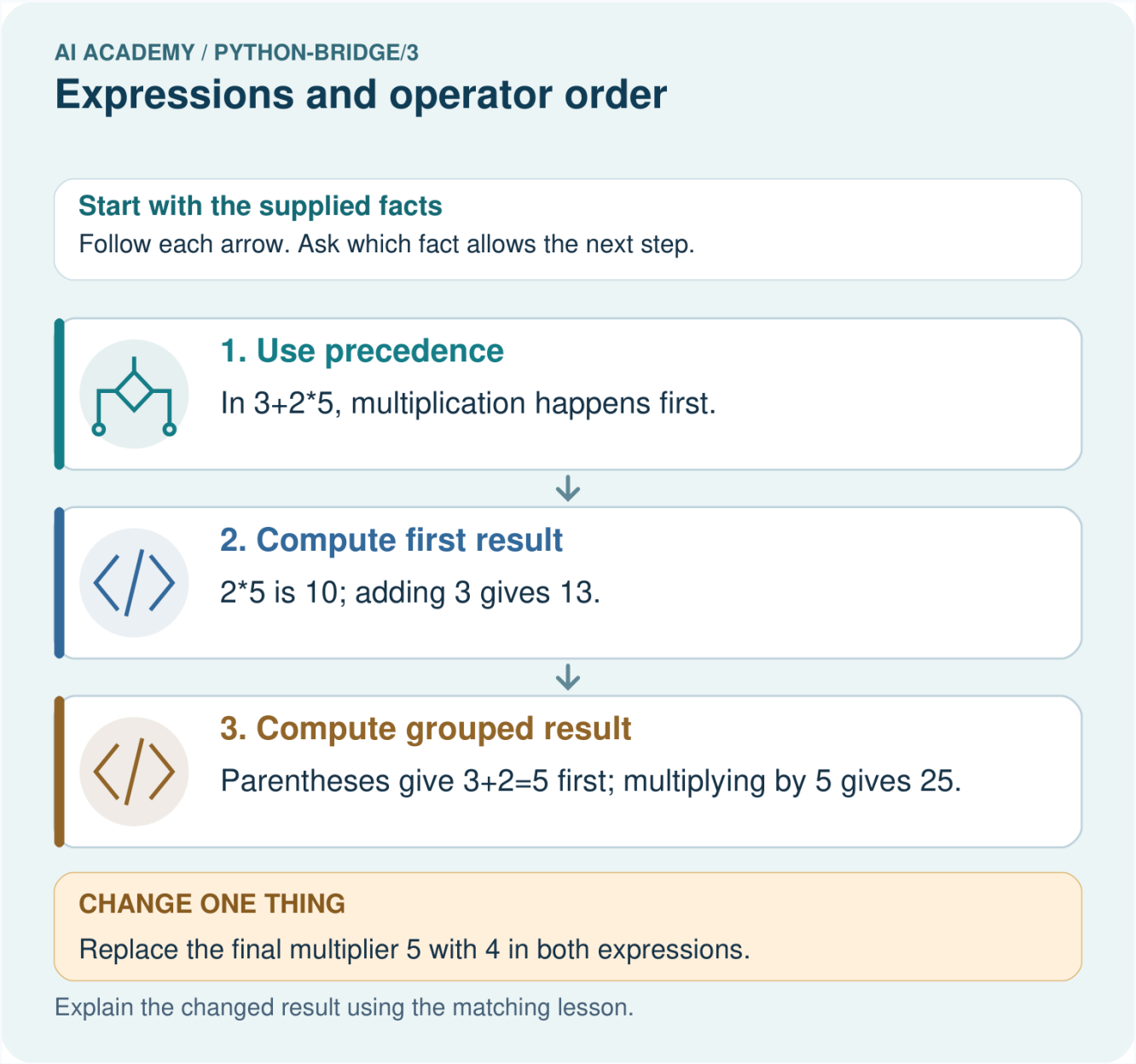
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Expressions and operator order

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the trace, then fade the help

> Collect predictions before execution or feedback.

Main completion task and key

Completion task: Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

Teacher check: (16 - 5) - 3 is 8; 16 - (5 - 3) is 14; (18 / 3) * 2 is 12.0. Record any observed discrepancy and check the saved **expression** rather than copying expected results.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Expression**s and **operator** order. Proposed explanation: "All operations are read left to right regardless of **precedence**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Establish grouping first; only operations at the same applicable **precedence** level group left to right here.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case key

Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Without grouping, multiply before adding: 13. With grouping, add before multiplying: 25. Changing the multiplier to 4 gives 11 and 20 respectively.

Changed-case reasoning: The first result is 11; the grouped result is 20. 3+2*4=3+8; (3+2)*4=5*4.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

▼ Support and alternative plan

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve numeric addition versus text joining.
7 to 20 minutes	Model packs and loose counters with the diagram; complete W1.
20 to 33 minutes	Read equal-precedence expressions from left to right and predict W2.
33 to 45 minutes	Run W2, compare results and collect the guided explanation.
Session 2 start to 8 minutes	Revisit one grouping disagreement with counters or brackets.
8 to 20 minutes	Model 17 counters in groups of five and complete W3.
20 to 35 minutes	Collect W4 independently, including the reconstruction check.
35 to 45 minutes	Complete W5 and the exit, recording input assumptions and units.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Colour the pack calculation and loose quantity separately, then use the same grouping in code. Provide an operation legend without doing W4.

Extension: devise two different **expression**s that agree for one **input** and differ for another, and explain the testing consequence.

▼ **Earlier wording, alternatives and picture transcripts**

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Preserve expected, observed and paper-trace records.

Supplied case materials and checked guidance

W2 outputs 8, 14, 12.0 from stated grouping. 17/5=3.4,17//5=3,17%5=2; whole-count interpretation assumes nonnegative counters and positive group size. Zero divisor raises ZeroDivisionError. Selected independent outputs 4, 18, then 3.4 3 2; grouped remainder (17+5)%5=2. Explain precedence and prior exposure rather than award independence for a memorised number. Earlier W4 task has its own quantities.

Diagnose misconceptions

All operations are read left to right regardless of **precedence**: Establish grouping first; only operations at the same applicable **precedence** level group left to right here.

Parentheses never change a value: Compare the two packs **expression**s and their different stories.

Ordinary division always returns an integer: / produces a float for these integer operands.

A **remainder** can be as large as the group size: For this positive whole-group case, another complete group would then be possible.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected bridge transfer task

> If a learner has seen this exact case, use a fresh parallel input and record exposure/support.

Selected independent task and checked key

Selected case: Predict each line. Explain which calculation groups addition before division. What would (17+5)%5 return, and how do /, // and % differ for the supplied positive integers?

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

The first line is4 because22//5=4. The second is18 because5//5 is evaluated before adding17. The final line is3.4 3 2: ordinary division, whole quotient and **remainder** for17 divided by5. (17+5)%5 is2.

Parentheses choose the grouped calculation; they are not optional when that meaning is intended.

Judge trace and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 State the intended grouping

Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

3 * 8 + 2 or (3 * 8) + 2 gives 26. 3 * (8 + 2) gives 30 and adds two to each of three packs, not two once.

W2 Predict then test expressions

Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

(16 - 5) - 3 is 8; 16 - (5 - 3) is 14; (18 / 3) * 2 is 12.0. Record any observed discrepancy and check the saved **expression** rather than copying expected results.

W3 Explain whole groups

For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.

/ gives 3.4; // gives 3; % gives 2. The reconstruction is 3 * 5 + 2 = 17, with **remainder** 2 less than group size 5. Only three complete groups can be made.

W5 Check a limit

Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

A zero-size group cannot define how many groups consume the counters; Python division by zero raises ZeroDivisionError.

The physical task uses whole nonnegative counters and a positive whole group size, so the **remainder** is between zero and one less than group size.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

In 18 - 4 * 3 multiplication groups first, giving 18 - 12 = 6. In (18 - 4) * 3 the **parentheses** give 14 * 3 = 42.

Exit: 23 // 4 gives 5 trays and 23 % 4 gives 3 tiles; 5 * 4 + 3 = 23. / gives 5.75, a ratio rather than both requested whole-number counts.

W1 State the intended grouping

W2 Predict then test expressions

W3 Explain whole groups

W4 Translate a new situation independently

Four trays hold six tiles each and five tiles are loose. Write and evaluate the total **expression**. Then pack the total into complete groups of four and find the **remainder**. Explain why 4 * (6 + 5) is wrong for this story.

4 * 6 + 5 gives 29 tiles; 29 // 4 gives 7 groups and 29 % 4 gives 1 left. 7 * 4 + 1 reconstructs 29. 4 * (6 + 5) gives 44, adding five to every tray instead of once.

W5 Check a limit

Independent assessment and fresh retry

W4 needs 29, 7 and 1 with the meaning of the extra five and a reconstruction. After support, retry three trays of seven plus four loose: 25 total; groups of six give 4 complete groups and 1 left.

Require an **expression** and explanation before a device result.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Keep the portfolio, support and delayed check

> Track code execution separately from a correct paper trace.

Project evidence

Add a labelled **expression**s and **operator** order **evidence** sheet to Python readiness portfolio. Show how you can evaluate numeric **expression**s using python's grouping rules.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Support and selected assessment guidance

Colour the pack calculation and loose quantity separately, then use the same grouping in code. Provide an operation legend without doing W4.

Extension: devise two different **expression**s that agree for one **input** and differ for another, and explain the testing consequence.

Selected independent answer: The first line is4 because22//5=4. The second is18 because5//5 is evaluated before adding17. The final line is3.4 3 2: ordinary division, whole quotient and remainder for17 divided by5. (17+5)%5 is2. Parentheses choose the grouped calculation; they are not optional when that meaning is intended.

Keep earlier W4 transfer as additional practice; do not substitute its numbers or key for this selected task. Record prior exposure and use a fresh parallel case after support.

Delayed return

Suggested return points in this level: python-bridge/4, python-bridge/6, python-bridge/9. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to the next topic

Week 4 introduces comparisons and Boolean conditions that express whether requirements hold.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the teaching cycle.

Planning and assessment reference

Teaching target: Evaluate numeric **expression**s using Python's grouping rules

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

Reasoned practice answer: (16 - 5) - 3 is 8; 16 - (5 - 3) is 14; (18 / 3) * 2 is 12.0. Record any observed discrepancy and check the saved **expression** rather than copying expected results.

Check every learner before discussion: Use the worked case for **Expression**s and **operator** order. Proposed explanation: “All operations are read left to right regardless of **precedence**”.

Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Establish grouping first; only operations at the same applicable **precedence** level group left to right here.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace the final multiplier 5 with 4 in both **expression**s. Require a **prediction** and a reason before comparing. Expected result: The first result is 11; the grouped result is 20. Reason: 3+2*4=3+8; (3+2)*4=5*4.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep individual first attempts separate from supported rehearsal.

Feedback and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

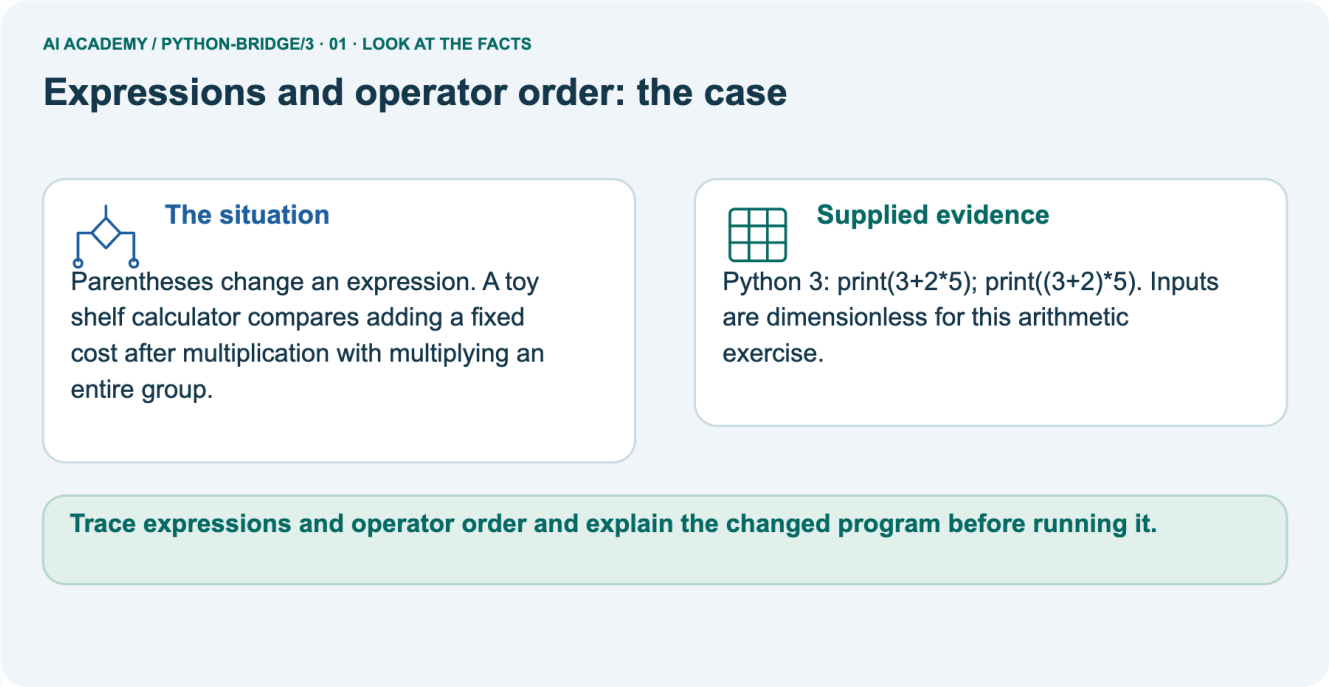
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Use the relevant picture for a specific reasoning move.

01. Expressions and operator order: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

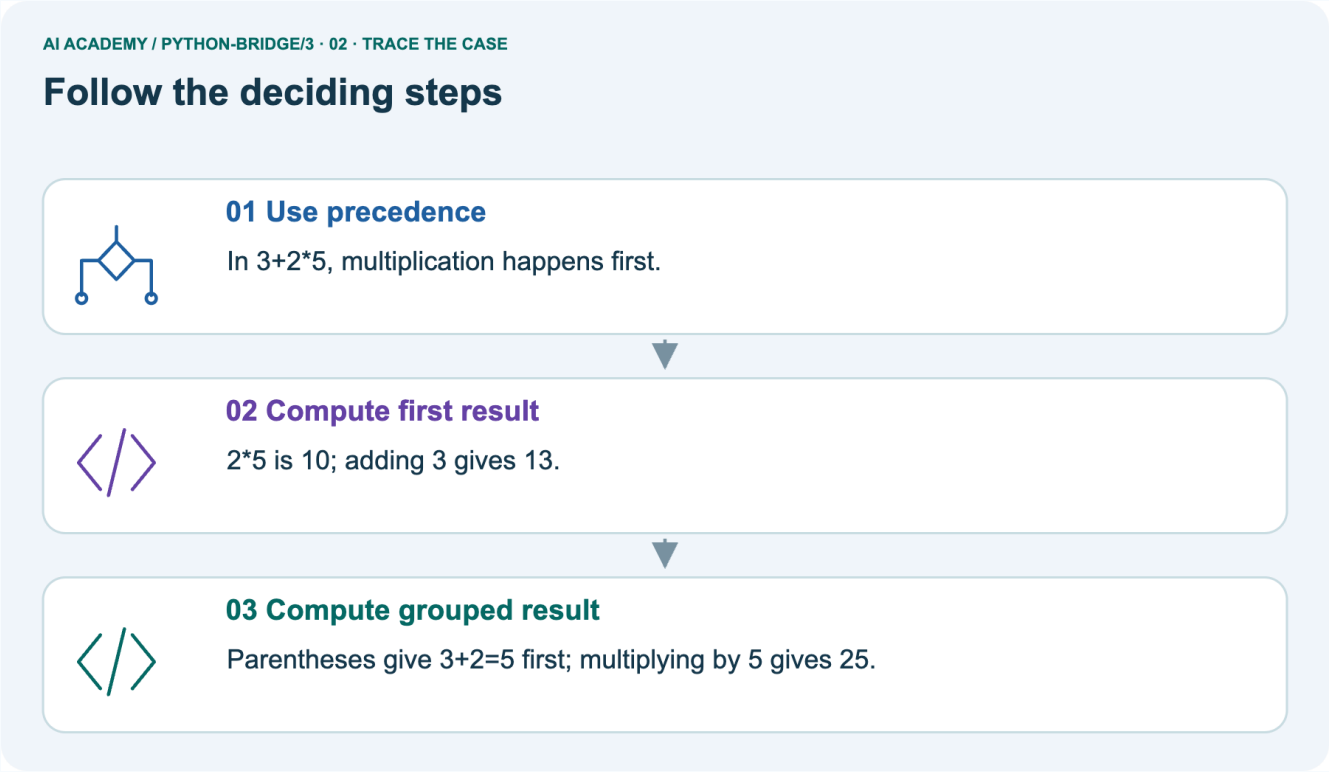
Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Supplied facts: Python 3: print(3+2*5); print((3+2)*5).

Inputs are dimensionless for this arithmetic exercise.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



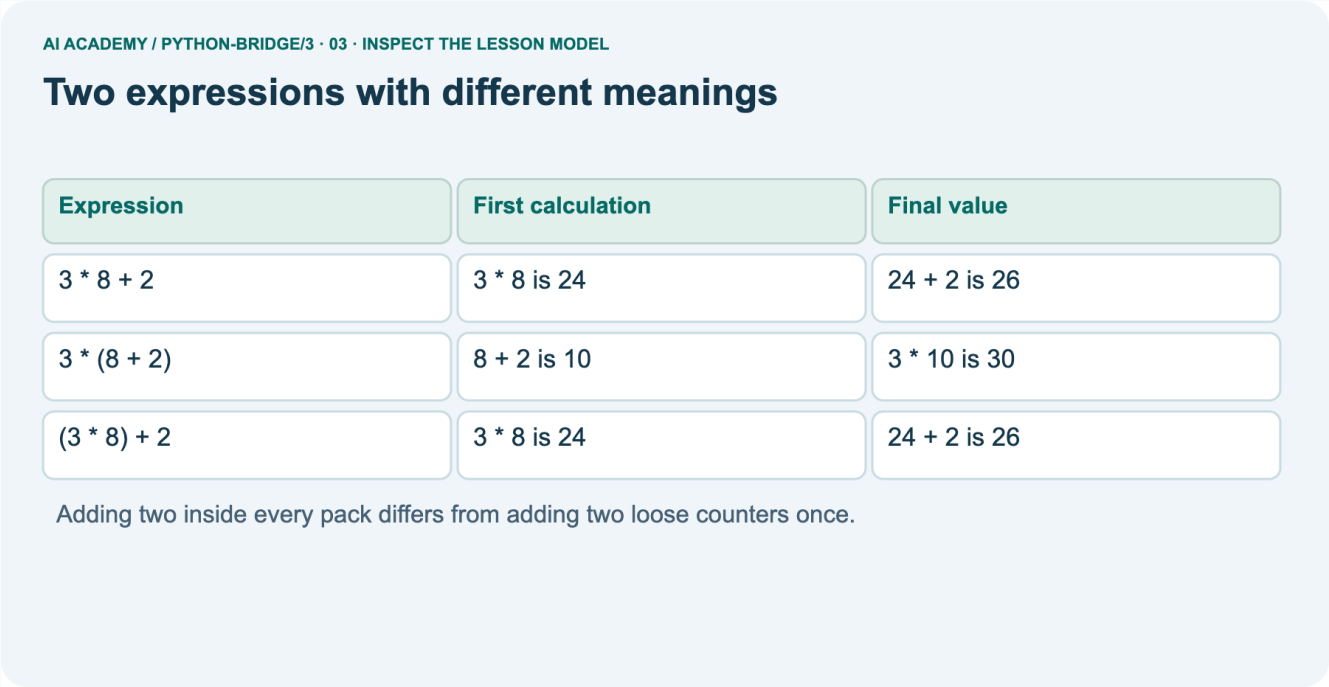
Follow the labelled picture with the matching explanation. Open the vector diagram

Use **precedence**: In 3+2*5, multiplication happens first. → Compute first result: 2*5 is 10; adding 3 gives 13. → Compute grouped result: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two expressions with different meanings



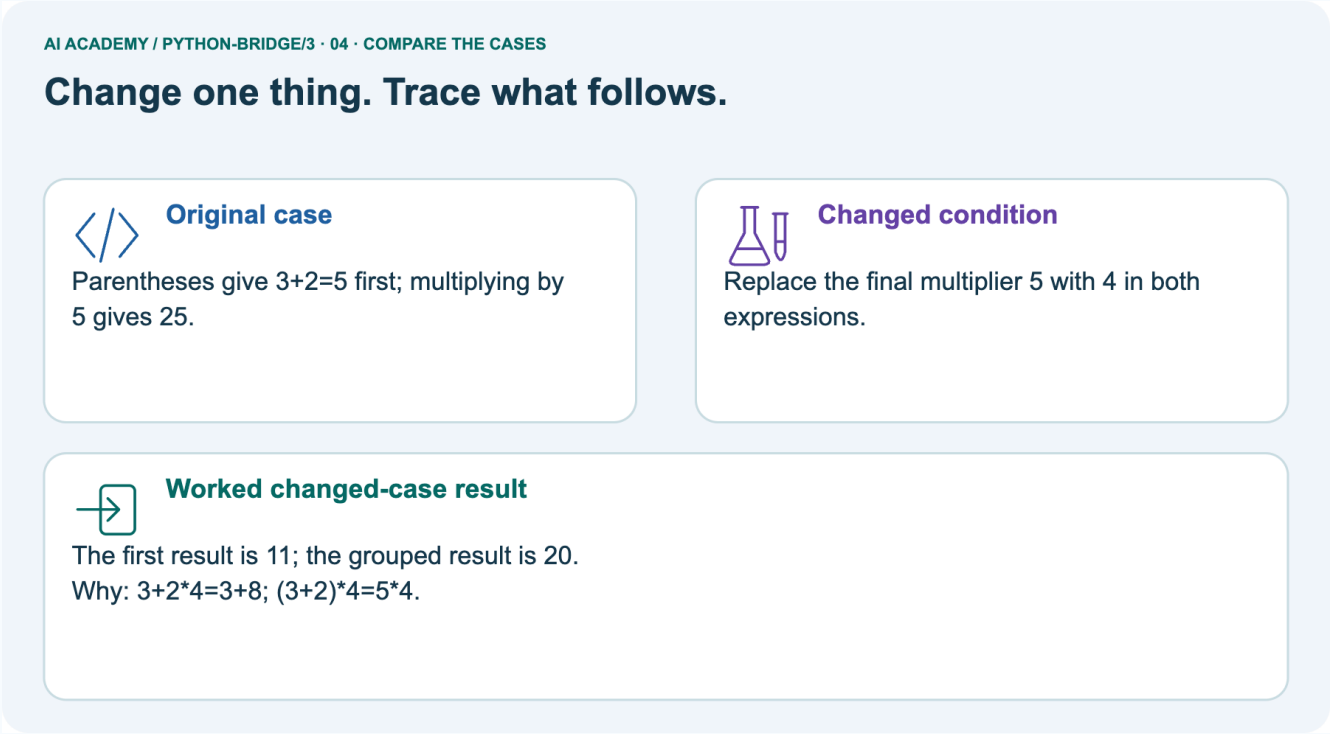
Follow the labelled picture with the matching explanation. Open the vector diagram

Two **expression**s with different meanings. 3 * 8 + 2 | 3 * 8 is 24 | 24 + 2 is 26. 3 * (8 + 2) | 8 + 2 is 10 | 3 * 10 is 30. (3 * 8) + 2 | 3 * 8 is 24 | 24 + 2 is 26

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. Changed condition: Replace the final multiplier 5 with 4 in both **expression**s. Expected result: The first result is 11; the grouped result is 20.

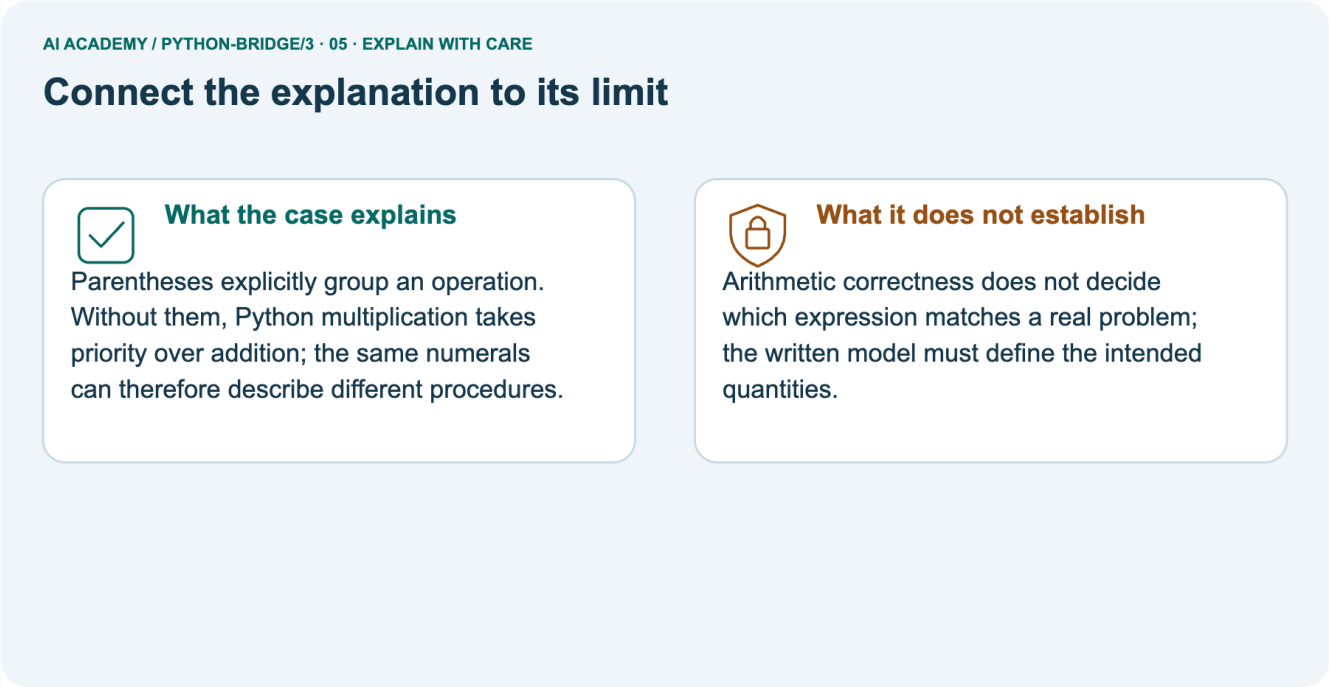
Reason: 3+2*4=3+8; (3+2)*4=5*4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 3+2*4=3+8; (3+2)*4=5*4.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

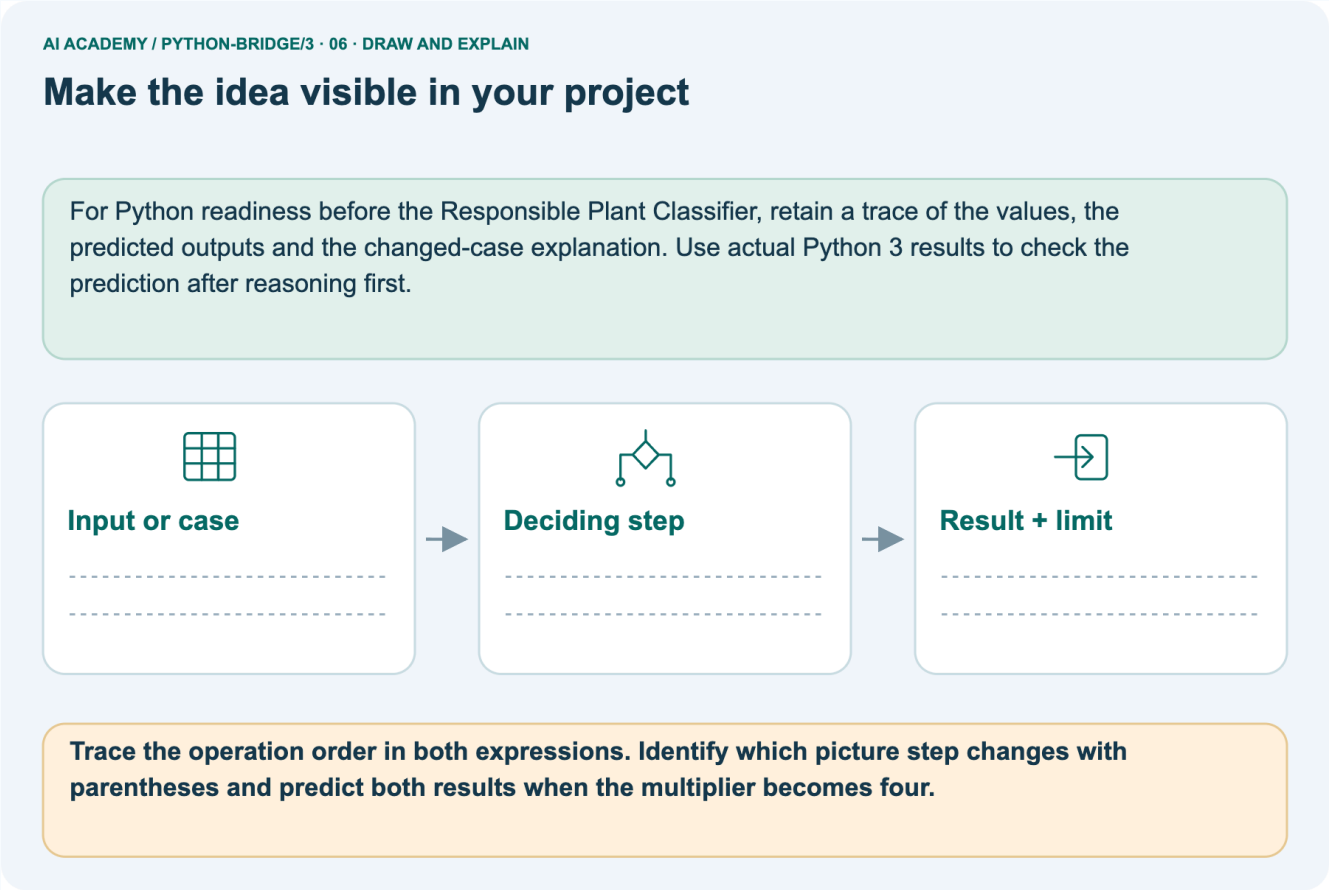
Limit: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Trace the operation order in both **expression**s.

Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Without grouping, multiply before adding: 13. With grouping, add before multiplying: 25. Changing the multiplier to 4 gives 11 and 20 respectively.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Expressions and operator order: the case

02. Follow the deciding steps

03. Two expressions with different meanings

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.